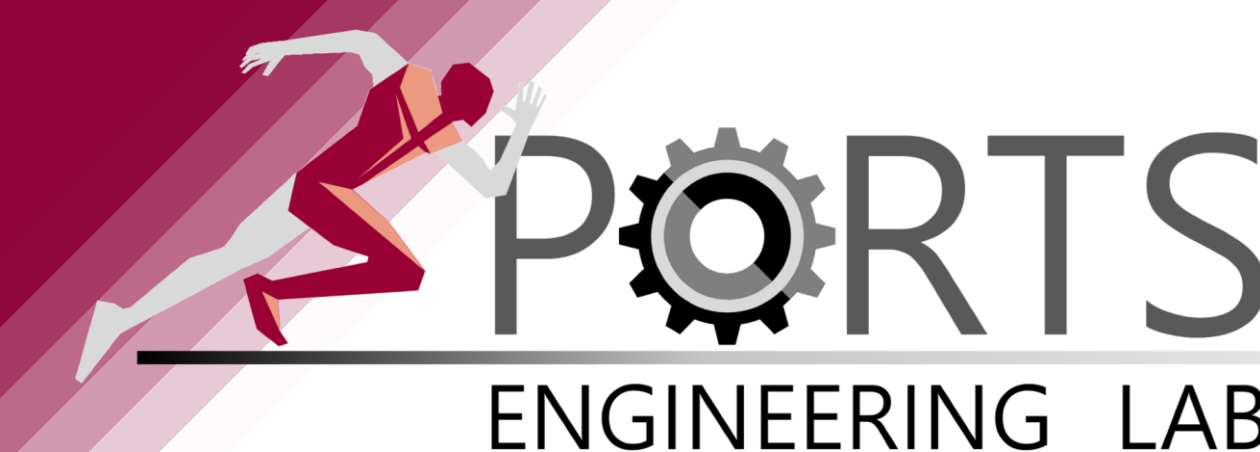




CHIHYEONG LEE
chi0412@snu.ac.kr
Advisor: Dr. Jooeun Ahn

Looking for:
POSTDOC. POSITION

Evaluating, Understanding, and Predicting Human Movement



EDUCATION

Seoul National University, Sports Engineering Lab

- Ph.D. in Sports Science (*Expected Feb 2027*)
- M.S. in Sports Science, 2023

University of Houston, SMART Lab

- Visiting Researcher, 2023

RESEARCH PHILOSOPHY

"My goal is to integrate biomechanics, wearable robotics, and digital human simulation to evaluate, understand and predict human movement."

PUBLISH

- **PAPERS:** 3 papers (3 first-author) + 1 under review
- **CONFERENCES:** 6 abstracts, 3 full papers

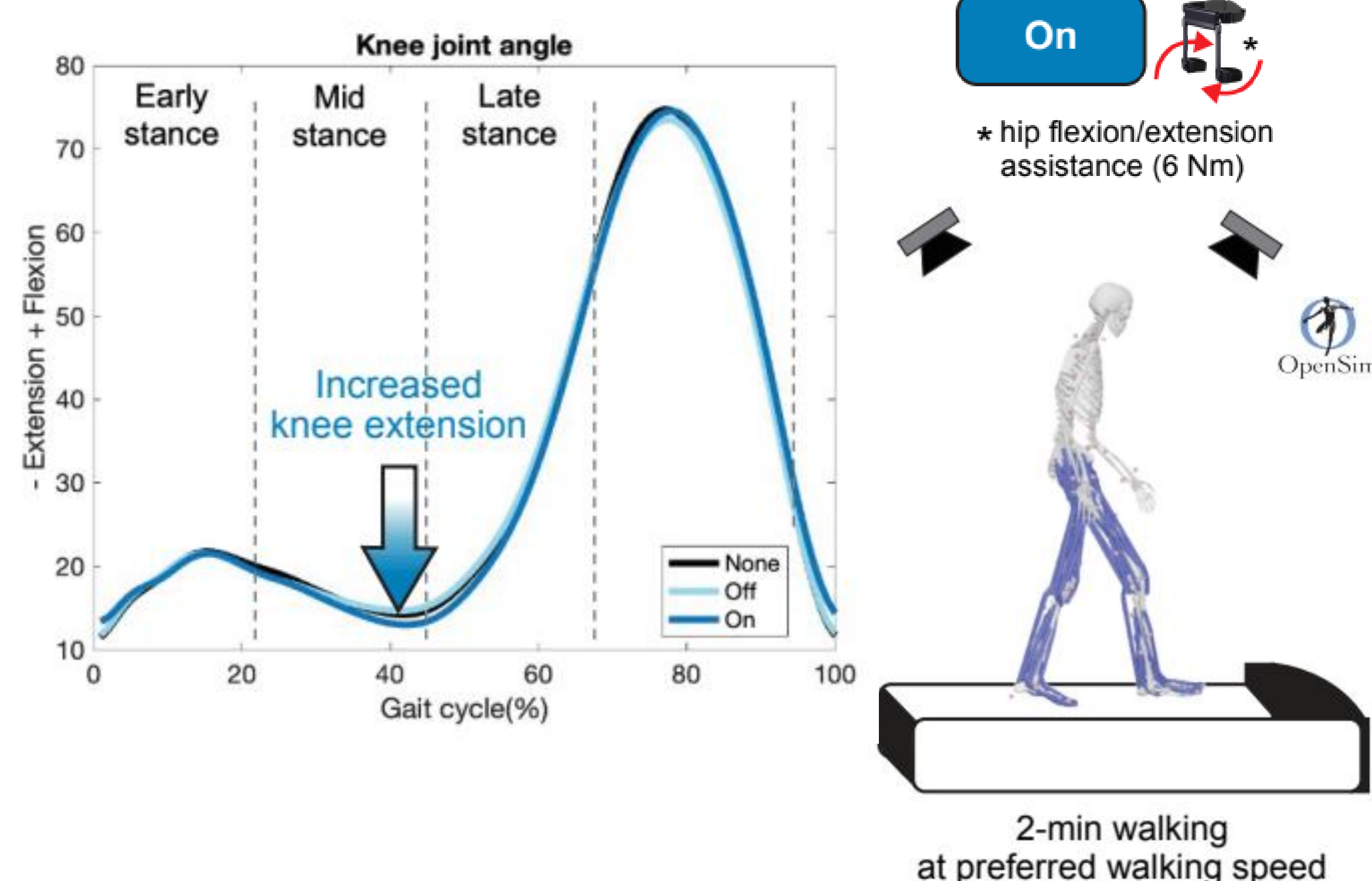
HUMAN-ROBOT INTERACTION

RAL (2025)



*"The proposed fully soft lift-assist wearable suit showed the potential to **reduce muscle activation and metabolic cost** during lifting tasks."*

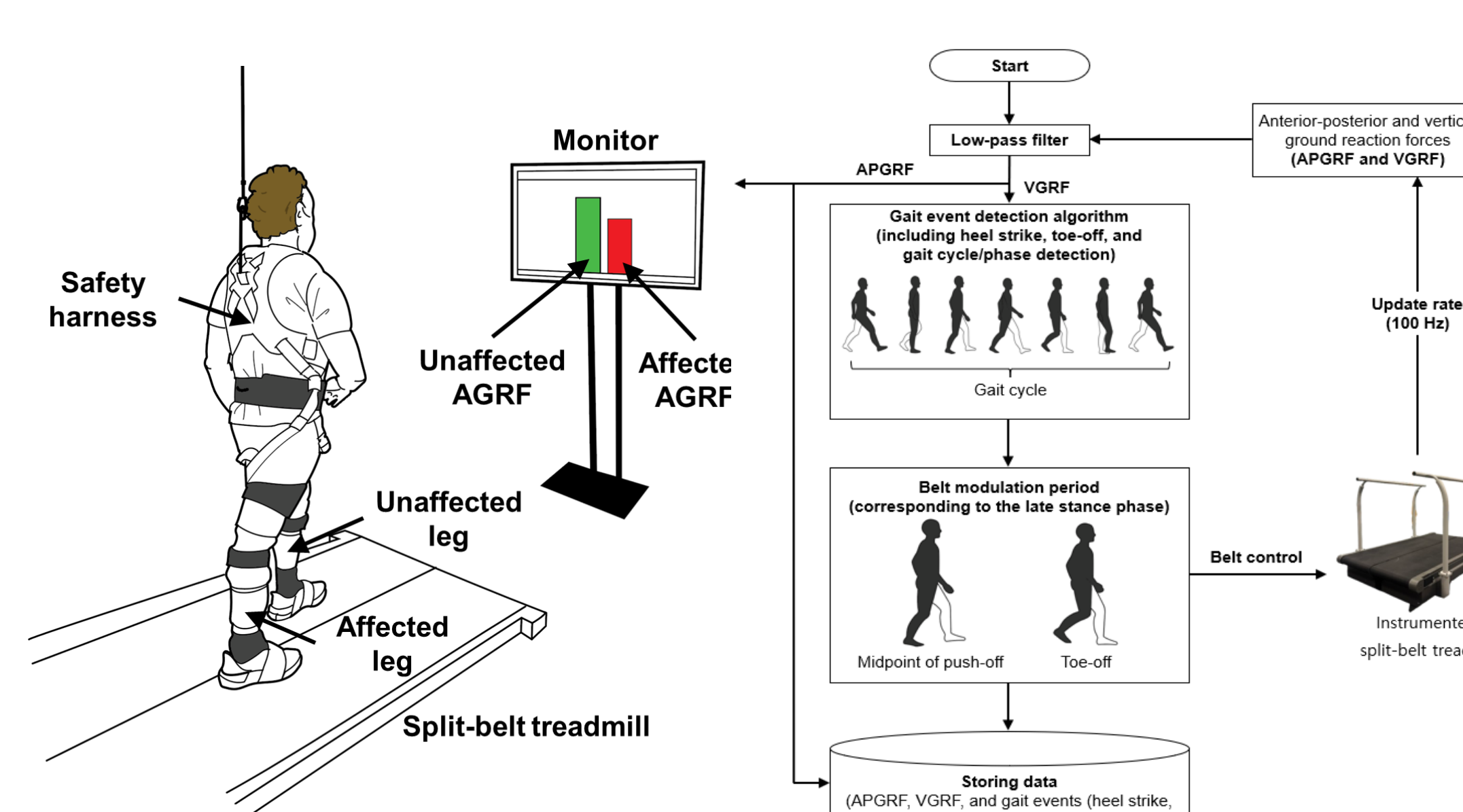
EMBC (2025)



*"Vector coding revealed that a **hip assist robot** altered **hip-knee coordination**, particularly during the mid-to-late stance phase."*

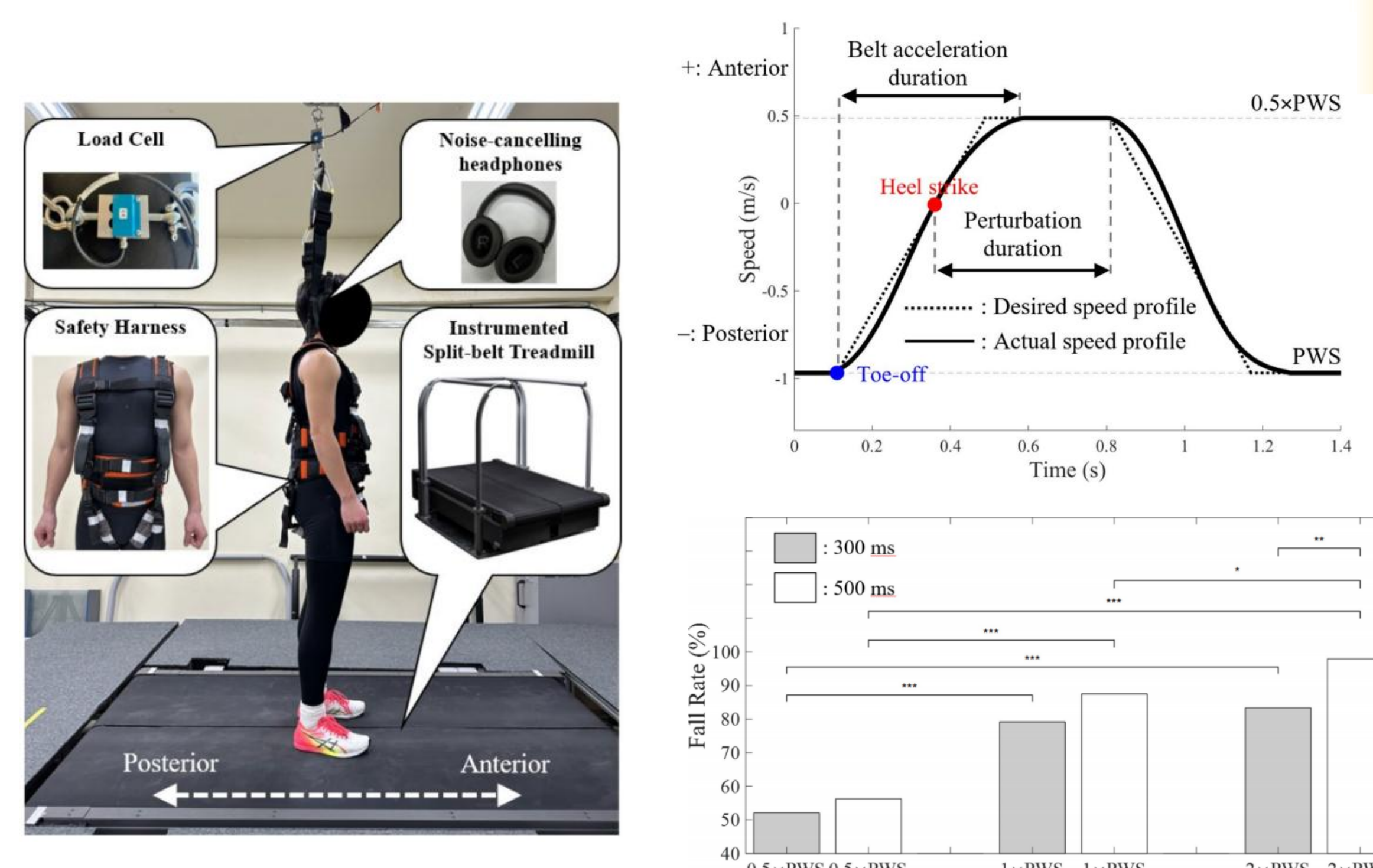
HUMAN RESPONSE ANALYSIS

JNER (2026, under review)



*"We developed a **real-time post-stroke gait training system** by customizing an instrumented split-belt treadmill. **Visual biofeedback and adaptive belt deceleration synergistically enhanced gait propulsion.**"*

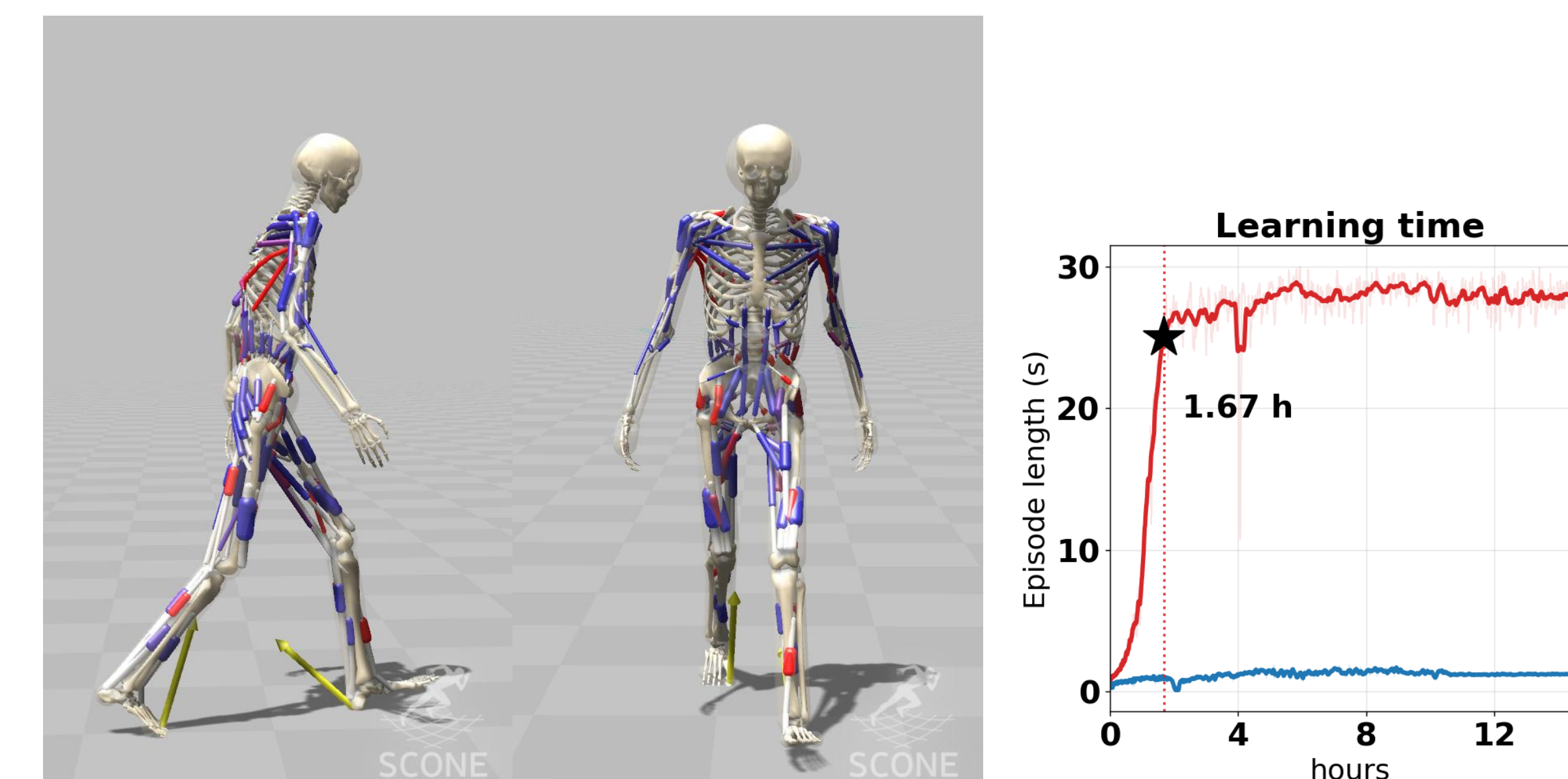
Scientific Reports (2025)



*"We customized an instrumented split-belt treadmill to deliver **real-time slip perturbations** and establish **design criteria for fall-inducing experiments.**"*

HUMAN MOVEMENT PREDICTION

WCB (2026)

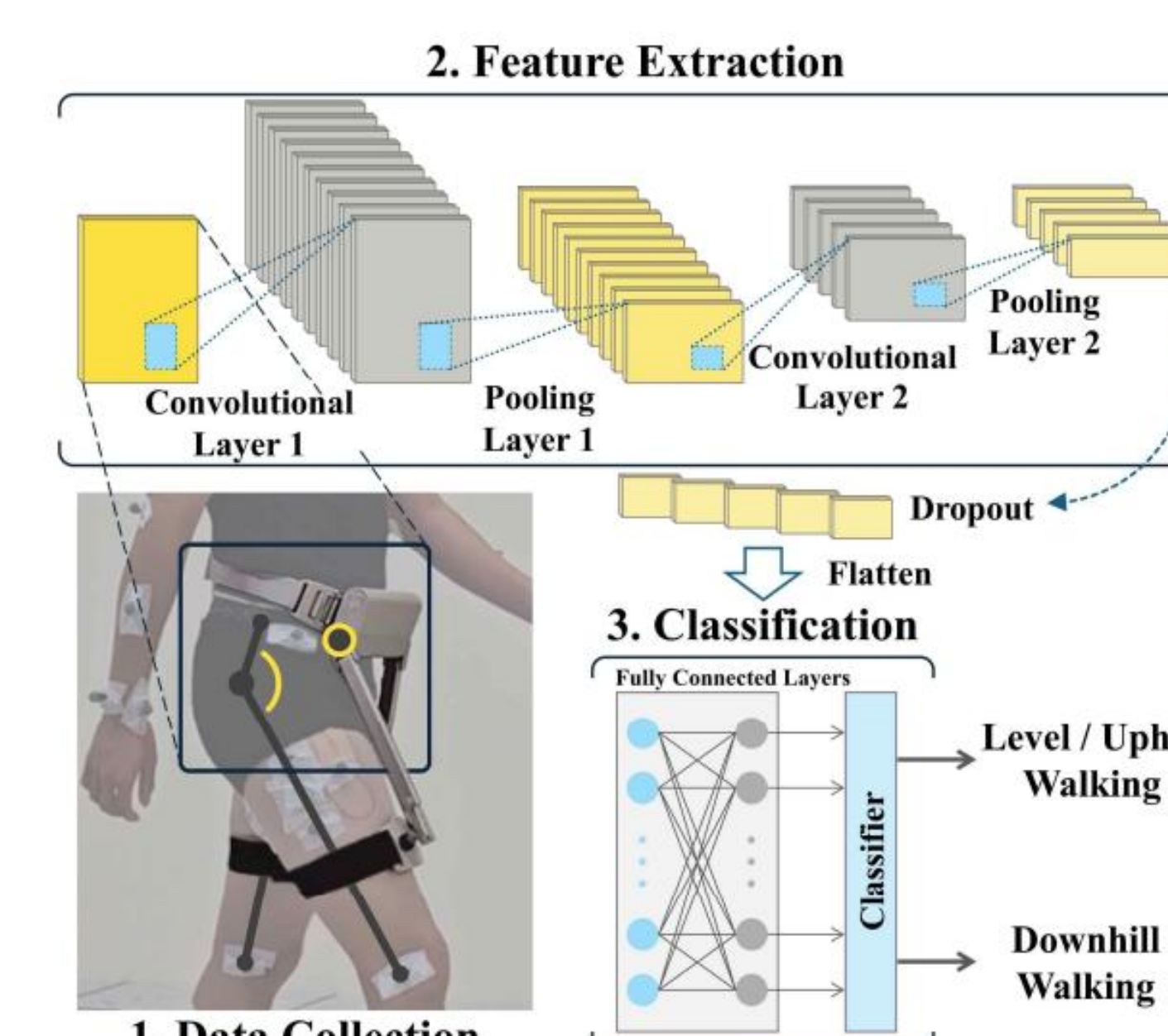


"Reference Configuration Imitation Learning (RCIL) achieved fast and stable walking by learning reference targets rather than direct muscle activations."

ISBS (2023)

"Machine learning enabled accurate and real-time detection of backward slips for safer gait-assistive wearable robots."

ASB (2025)



*"A **deep learning** approach enabled **accurate and real-time classification of downhill walking** for adaptive gait-assistive wearable robots."*

SKILLS

Experimental Device

OptiTrack®

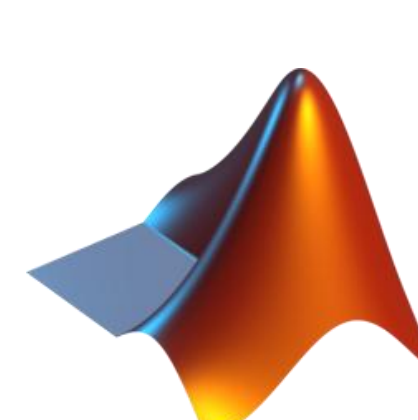
AMTI
FORCE AND MOTION

BERTEC



+ Custom Software

Programming



Visual 3D

OpenSim



Musculoskeletal Simulation

HYFYDY
HIGH FIDELITY DYNAMICS

LET'S STAY IN TOUCH!

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SPORTS
ENGINEERING LAB

